

CENG381 Stochastic Processes

Irreducibility and Periodicity - Answer Key 1

Q1 Solution

a) Transition diagram:

$0 \leftrightarrow 1 \leftrightarrow 2 \leftrightarrow 3$ (a path graph with bidirectional edges)

$0 \xrightarrow{1} 1 \xrightarrow{0.5} 0$ and $1 \xrightarrow{0.5} 2$

$2 \xrightarrow{0.5} 1$ and $2 \xrightarrow{0.5} 3$

$3 \xrightarrow{1} 2$

b) Irreducibility:

From 0: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ (reachable in 3 steps)

From 3: $3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ (reachable in 3 steps)

Every state can reach every other state $\Rightarrow$ the chain is IRREDUCIBLE.

c) Periods:

State 0: can return to 0 in 2 steps ($0 \rightarrow 1 \rightarrow 0$), 4 steps, 6 steps, ...

$$T(0) = \{2, 4, 6, \dots\} \Rightarrow d(0) = \gcd(2, 4, 6, \dots) = 2$$

State 1: can return to 1 in 2 steps ($1 \rightarrow 0 \rightarrow 1$ or $1 \rightarrow 2 \rightarrow 1$)

$$T(1) = \{2, 4, 6, \dots\} \Rightarrow d(1) = 2$$

State 2: same argument $\Rightarrow d(2) = 2$

State 3: can return to 3 in 2 steps ($3 \rightarrow 2 \rightarrow 3$)

$$T(3) = \{2, 4, 6, \dots\} \Rightarrow d(3) = 2$$

All states have period 2. The chain is PERIODIC (not aperiodic).

This means the chain alternates between even and odd states and does NOT converge to the stationary distribution.

Q2 Solution

a) Transition matrix for random walk on 4-cycle:

	0	1	2	3
0	[0.0	0.5	0.0	0.5]
1	[0.5	0.0	0.5	0.0]
2	[0.0	0.5	0.0	0.5]
3	[0.5	0.0	0.5	0.0]

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b) P^2 block structure:

$$\begin{aligned} P^2 \text{ row 0: } P[0][.] * P &= (0.5*P[1] + 0.5*P[3]) \\ &= 0.5*(0.5, 0, 0.5, 0) + 0.5*(0.5, 0, 0.5, 0) = (0.5, 0, 0.5, 0) \end{aligned}$$

$$P^2 \text{ row 1: } (0, 0.5, 0, 0.5)$$

$$P^2 \text{ row 2: } (0.5, 0, 0.5, 0)$$

$$P^2 \text{ row 3: } (0, 0.5, 0, 0.5)$$

P^2 has zero entries wherever P had non-zero entries $\Rightarrow$ the even states $\{0, 2\}$ and odd states $\{1, 3\}$ never communicate in 2 steps.

This confirms period $d = 2$ for all states (returns only happen in even numbers of steps).

c) Lazy chain $Q = (1/2)I + (1/2)P$:

	0	1	2	3
0	[0.50	0.25	0.00	0.25]
1	[0.25	0.50	0.25	0.00]
2	[0.00	0.25	0.50	0.25]
3	[0.25	0.00	0.25	0.50]

Aperiodicity of Q : State 0 has $P(\text{return to 0 in 1 step}) = Q[0][0] = 0.5 > 0$.

So $T(0)$ contains 1, and gcd of any set containing 1 equals 1.

Therefore $d(0) = 1$, and by irreducibility all states have period 1.

The lazy chain is aperiodic. (OK)

Q3 Solution

b) Communicating classes of the web graph:

From page 1: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ (cycle: 1 reaches 2, 3, 4)

From page 2: $2 \rightarrow 1 \rightarrow 2$ and $2 \rightarrow 3 \rightarrow 4 \rightarrow 1 \rightarrow 2$ (2 reaches 1, 3, 4)

From page 3: $3 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 3$ (3 reaches 4, 1, 2)

From page 4: $4 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ (4 reaches 1, 2, 3)

All 4 pages form one communicating class $\Rightarrow$ the chain IS irreducible.

However, it has period 4 (can only return in multiples of 4 steps).

So it is irreducible but periodic.

c) PageRank fix with damping $d = 0.85$, $n = 4$:

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$$\begin{aligned}P' &= 0.15 \cdot (1/4) \cdot J + 0.85 \cdot P \\ &= 0.0375 \cdot J + 0.85 \cdot P\end{aligned}$$

Each entry of $P' = 0.0375 + 0.85 \cdot P[i][j]$

For example, $P'[1][1] = 0.0375 + 0.85 \cdot 0.0 = 0.0375$

$$P'[1][2] = 0.0375 + 0.85 \cdot 1.0 = 0.8875$$

$$P'[1][3] = 0.0375 + 0.85 \cdot 0.0 = 0.0375$$

$$P'[1][4] = 0.0375 + 0.85 \cdot 0.0 = 0.0375$$

Why irreducible: every entry of P' is > 0 (since $0.0375 > 0$), so $P'^1(i,j) > 0$ for ALL pairs (i,j) . Every state reaches every other state in exactly one step.

Why aperiodic: $P'[i][i] \geq 0.0375 > 0$, so $T(i)$ contains 1, meaning $\gcd = 1$ for all states $\Rightarrow$ aperiodic.

Therefore P' is both irreducible and aperiodic, guaranteeing convergence of the PageRank vector to a unique stationary distribution.